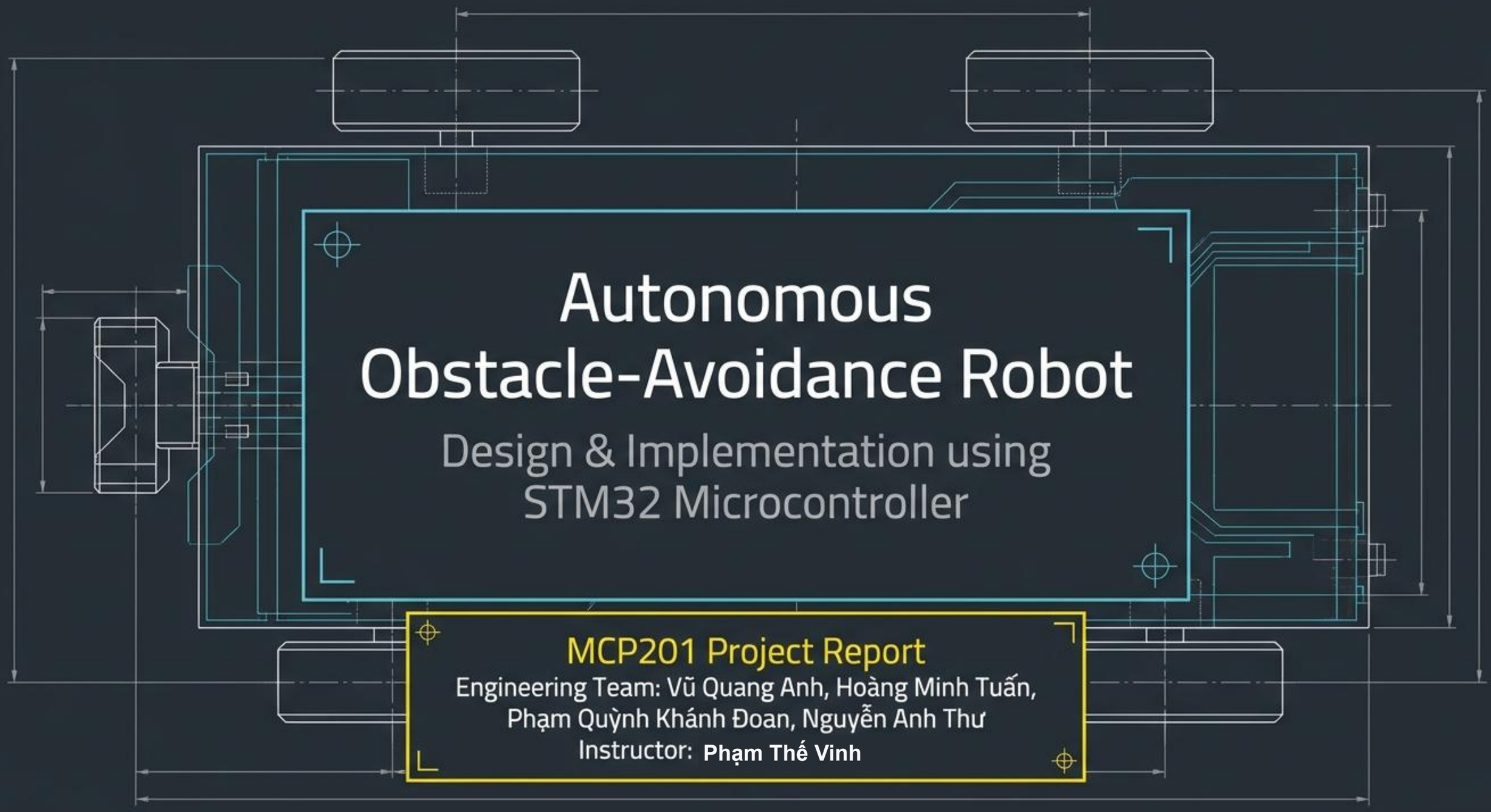


Autonomous Obstacle-Avoidance Robot

Design & Implementation using STM32 Microcontroller

MCP201 Project Report

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Instructor: **Phạm Thế Vinh**

Project Scope and Core Engineering Objectives



Hardware Implementation

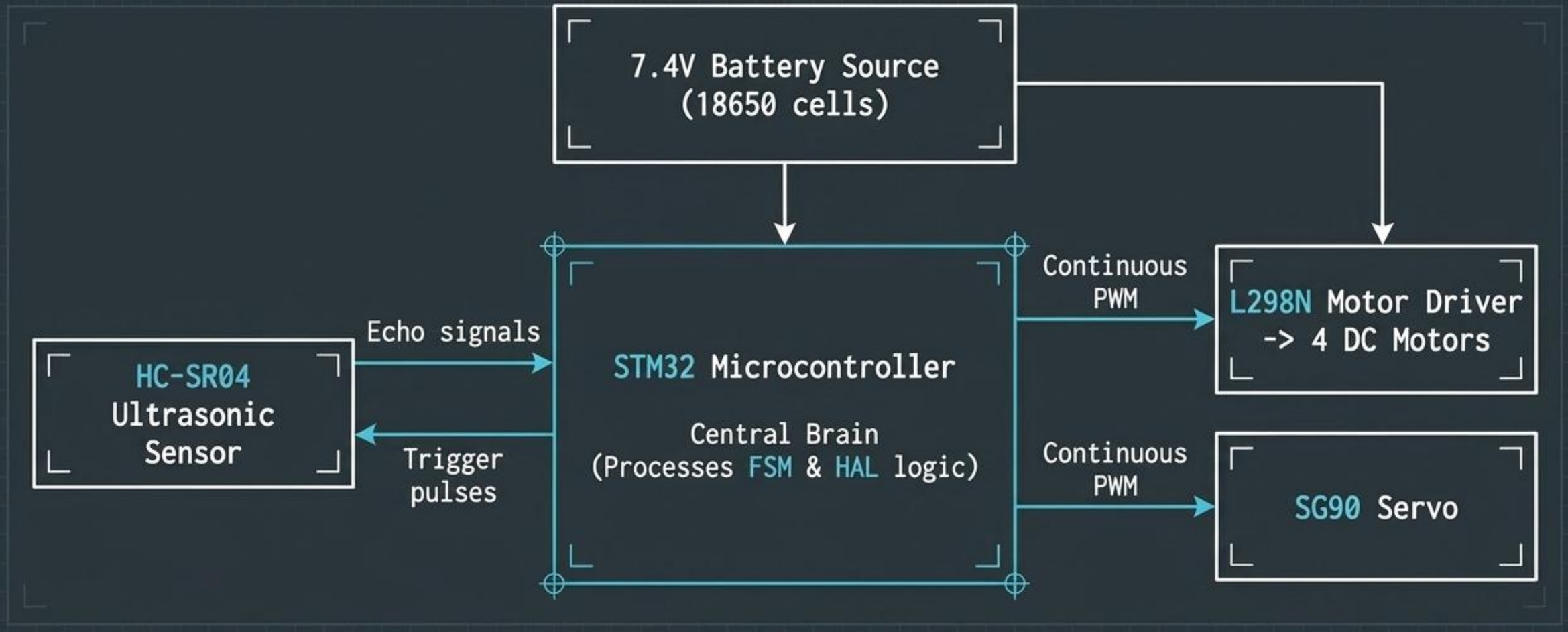
- Assemble a 4-wheel drive chassis.
- Integrate STM32 (Nucleo/Blue Pill) as the central controller.
- Drive 4 DC motors using an L298N H-bridge module.
- Expand sensor vision to 180° using an HC-SR04 ultrasonic sensor mounted on an SG90 Servo.
- Build an isolated power system to prevent electromagnetic interference.

{ } Software & Algorithms

- Configure STM32 peripherals (GPIO, Timer, PWM) via the HAL library.
- Measure distance with nanosecond precision using the Cortex-M DWT cycle counter.
- Filter acoustic noise using a 3-sample Median Filter algorithm.
- Implement a Finite State Machine (FSM) for non-blocking multitasking.

Project Limitations: Designed for indoor, flat surfaces. The audio warning module (DFPlayer) was excluded due to hardware failure; project focus was shifted entirely to optimizing core movement logic.

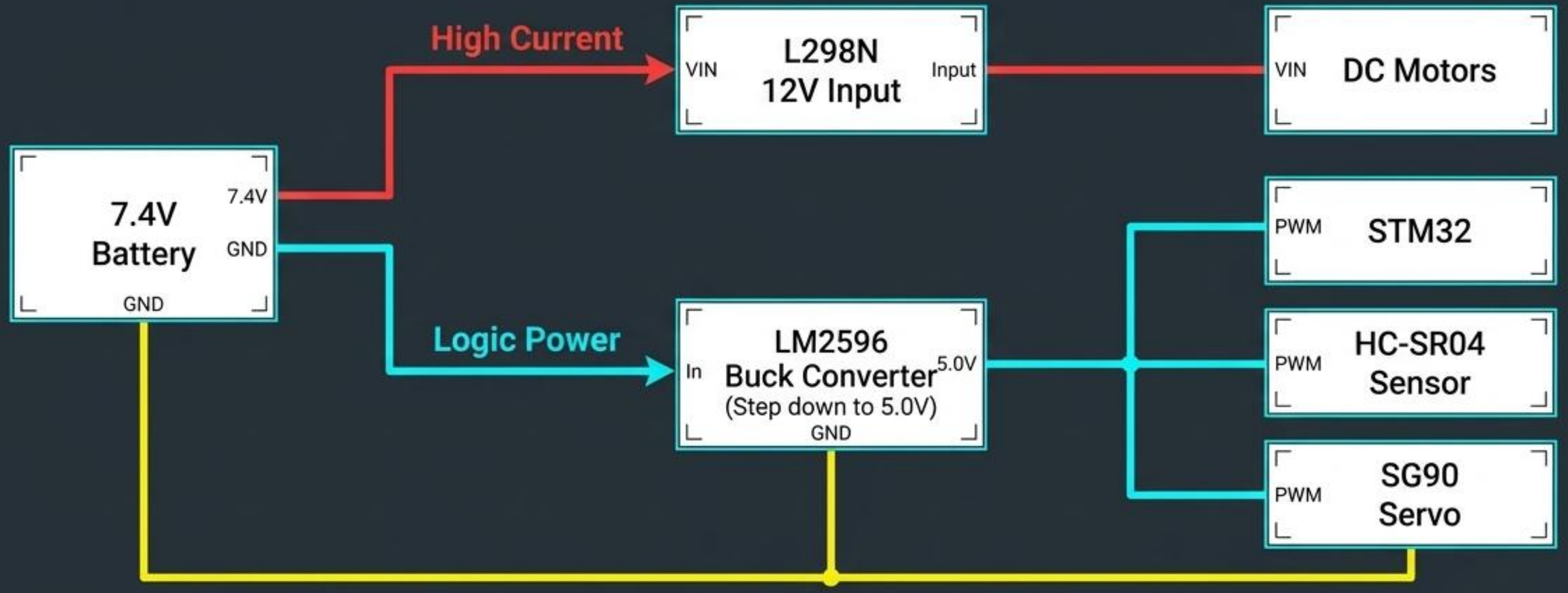
Overall System Architecture and Signal Flow



Key Takeaway: The system relies on precise timing for high-frequency inputs (sensor echoes) and continuous outputs (motor PWM). This hardware topology requires a strictly non-blocking software architecture to function.

Solving Hardware Interference Through Isolated Power Design

The Hardware Challenge: Voltage drops during DC motor startup cause STM32 resets and sensor misalignment.



⚠ Crucial Engineering Rule: Common Ground

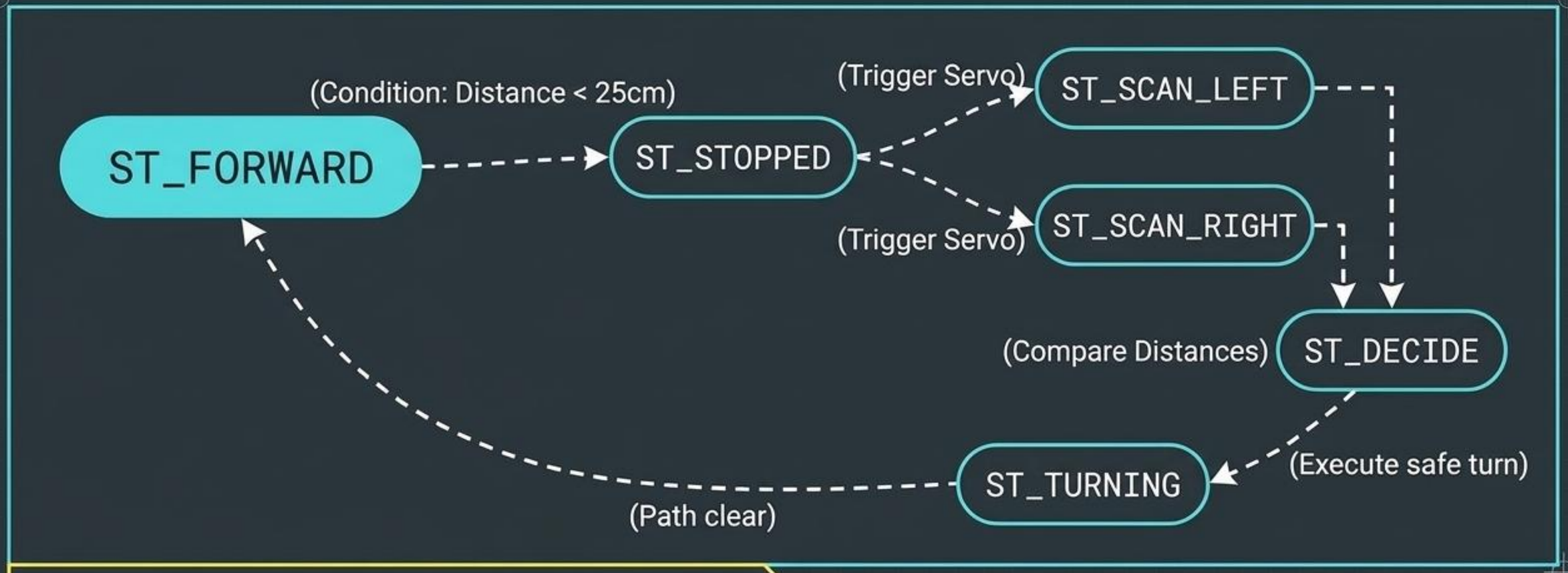
The L298N, LM2596, Sensor, Servo, and STM32 must share a single GND. This ensures all PWM and logic signals reference the exact same 0V baseline, eliminating signal drift.

Software Architecture: Avoiding the Infinite Loop Trap

Feature	Standard Approach	MCP201 Approach
Distance Measurement	`HAL_Delay()` (Blocks CPU entirely)	DWT Cycle Counter (Nanosecond precision, strictly non-blocking)
Navigation Logic	Linear top-to-bottom code	Finite State Machine (FSM) (Enables multitasking and instant environmental response)
Sensor Data Processing	Trusting single raw readings	Median Filter Array (Actively rejects sudden acoustic noise spikes)

Architectural Result: By completely eliminating HAL_Delay(), the STM32 can continuously evaluate incoming distances while simultaneously outputting uninterrupted PWM signals to the motors.

Decision Making via a Finite State Machine (FSM)



Code Implementation:
The setState() function acts as the centralized controller. When entering ST_STOPPED, motors halt immediately, but the CPU continues running the main loop to process the next scanning state without freezing.

Achieving Nanosecond Precision with the Cortex-M DWT



The Measurement Sequence:

1. Send a 10µs HIGH pulse to the TRIG pin.
2. Start the 32-bit `DWT->CYCCNT` register (ticks every single system clock cycle).
3. Wait for the ECHO pin to drop to LOW.
4. Calculate exact duration.

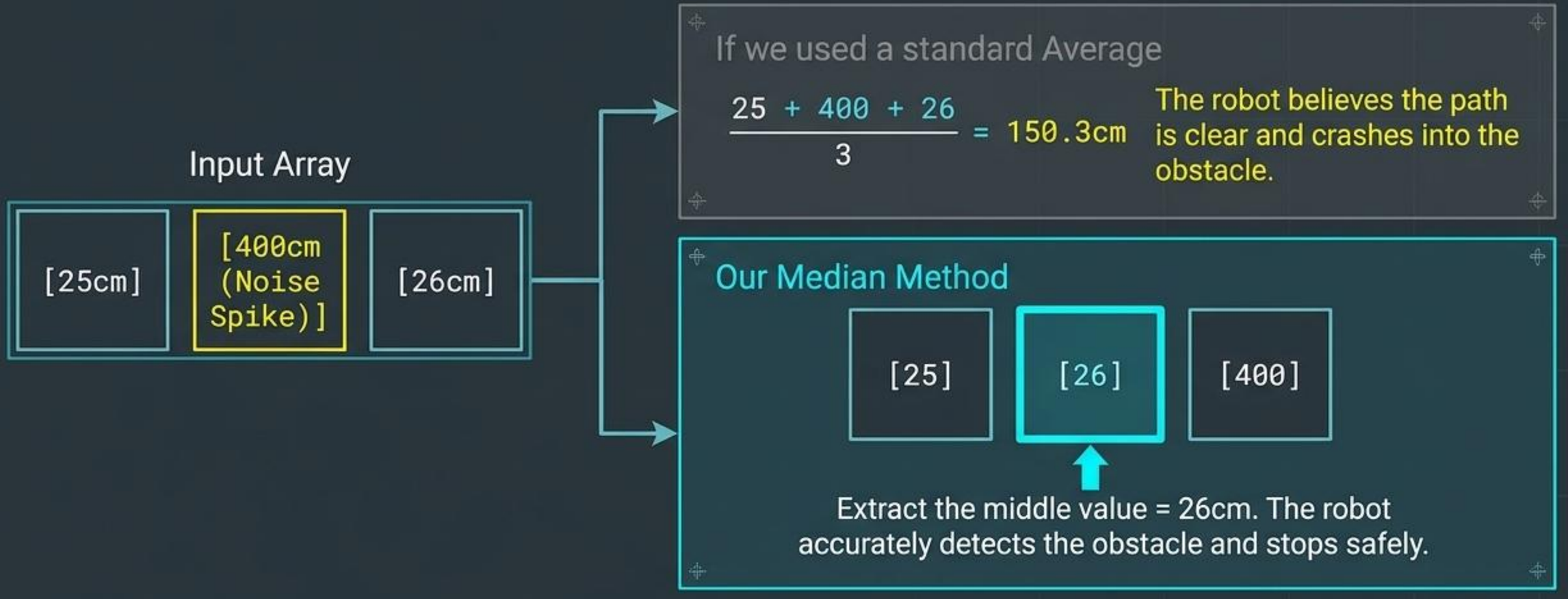
⚠ Safety Mechanism: Timeout Protection

The Risk: If an ultrasonic wave hits a soft object and scatters, the ECHO pin never drops, trapping the program in an infinite while loop.

The Fix: Implemented a 12,000µs threshold. If exceeded, the function forces a return of 999cm (max safe distance), preventing a total system crash.

Eliminating Sensor Noise with a Median Filter Array

The Environmental Problem: Uneven surfaces or acoustic reflections cause the HC-SR04 to occasionally return extreme, false measurements.



Execution: The system calls getDistanceNonBlocking every 50ms, constantly rolling this 3-sample array to ensure smooth, noise-free navigation.

Driving the L298N: Unifying Direction and PWM Speed

Hardware Interface



IN1 & IN2 Pins
Control motor rotation direction (Forward/Backward).

ENA Pin
Receives 1kHz PWM signal from Timer 1 to modulate speed.

Software Abstraction

Created a unified setLeftMotor(speed) function accepting values from -255 to 255.

If speed > 0: Set IN1/IN2 for Forward.

If speed < 0: Set IN1/IN2 for Backward, then convert speed to an absolute value.

Execution: Send the absolute value directly to __HAL_TIM_SET_COMPARE to set the Duty Cycle.

Result: Complex FSM commands are vastly simplified. A left turn is executed simply by calling setLeftMotor(-180) and setRightMotor(180).

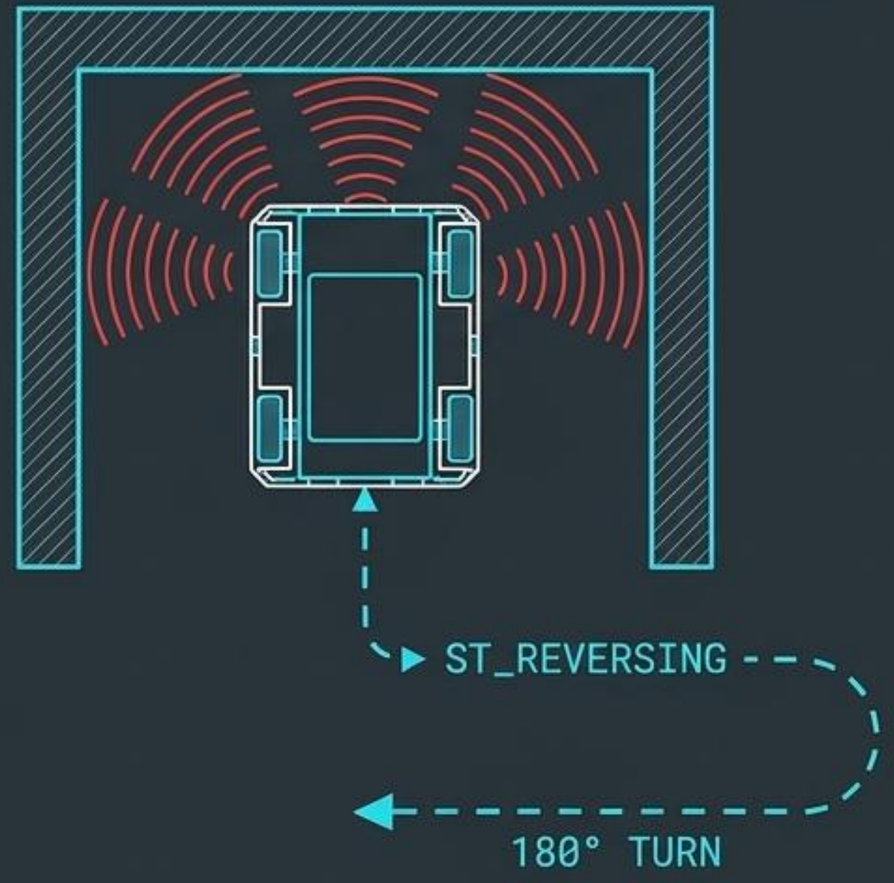
Empirical Validation and Testing Scenarios

Scenario	Expected Outcome	Result
Clear Path (>60cm)	Fast Forward.	✅ PASSED
Warning Zone (25-60cm)	Gradual deceleration.	✅ PASSED (1kHz PWM frequency provided highly smooth braking).
Obstacle (<25cm)	Immediate stop & scan.	✅ PASSED (No inertia drift observed upon braking).
Left Clear, Right Blocked	Turn left, resume.	✅ PASSED.

Sensor Accuracy Verified:

Live tracking via STM32CubeIDE (`dbg_dist` variable) confirmed distance error margins are strictly limited to ±1cm within the 10cm–100cm range.

Advanced Logic: Escaping Dead Ends and Narrow Corners



The Edge Case:
What happens when the robot drives into a narrow corner where the Front, Left, and Right scan distances are all **< 25cm**?

The Programmatic Solution:
- The FSM utilizes a `blockedCount` variable.
- If the `Servo` scans both left and right and fails to find a path wide enough, the counter exceeds its **safety threshold**.
- **Action Triggered:** The system forces a state transition to `ST_REVERSING` (backing straight out of the corner), immediately followed by a 180-degree turnaround sequence.

Result:
The autonomous system never gets permanently trapped in complex geometric environments.

Conclusion and Version 2.0 Roadmap

Project Conclusion: Successfully implemented an autonomous, multitasking robotic system utilizing strictly non-blocking STM32 HAL software and isolated hardware topology.

Phase 1: Closed-Loop PID Control

Add wheel Encoders to execute absolute straight-line movement, compensating for battery voltage fluctuations.

Phase 2: RTOS Migration

Transition the architecture from a standard FSM to FreeRTOS, dedicating independent CPU tasks for scanning, calculating, and driving.

Phase 3: Sensor Fusion

Integrate an IMU (MPU6050) for spatial orientation mapping and IR sensors to combine line-tracking with dynamic obstacle avoidance.